

Unique Perspective Brainstorming

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Your scope reducers:

- Mobile application
- Assisted Edge
- Repetitive Environments (University, Medical)
- Semantic Integration
- Privacy Features
- Map Updating
 - Federated Map Updating
- Dynamic FASTMAP

Ruled Out

- Plain Private Edge SLAM
- Plain Coordinate Mobile Offloading
- Semantic Assisted Edge SLAM ([EdgeSLAM 2](#) (2024))
- Cooperative & Semantic SLAM w/ Wifi ([TWC-SLAM](#))
- Semantic DUST3R/MASt3R/**Fast3R** ([Hier-SLAM++](#))
- Visual SLAM for Visually Impaired Navigation ([SELM-SLAM3](#))(2026)

Maybe

- Latest semantic edge SLAM is 2024
- Private Edge SLAM specific to Mobile
- Is Edge Assisted SLAM robot specific or phone inclusive?
- 2026 Semantic Assisted Edge SLAM
- AdaptSLAM within a repetitive environment
- Semantic DUST3R/MASt3R/**Fast3R** that is visual instead of gaussian
- FASTMAP incorporation into pipelines for mobile ready applications
- Semantic enabled Visual SLAM for Visually Impaired Navigation

Potential

- Edge-assisted DUST3R/MASt3R/**Fast3R**-style SLAM for mobile devices
 - The current mainstay is ORB-SLAM3 (2021)
 - This is because it is highly modular allow separate processing of components.
- Text aware implementation of DUST3R/MASt3R/**Fast3R**
- Anything privacy oriented on DUST3R/MASt3R/**Fast3R**
- persistent map updating for MASt3R/Fast3R SLAM
 - Almost certainly too challenging for me

Questions

- Why is ORBSLAM modular while other are not?
 - What modifications would be needed to do that?
- What monocular vision SLAMS ARE smartphone implementation tested?
- What would meaningful routing of inputs look like? It doesn't seem useful to simultaneously create two models....
 - Routing could be useful in the loop closure portion.
- Loop closure could be edge assisted. The could be routed depending on difficulty?
- Has anyone attempted to retrain SuperGlue/LightGlue to incorporate non visual tokens?